

An AI Clinical Diary: Integrating Patient Narratives and Sensor Data for Clinical Decision Support

User Study — Final Report

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KTH Royal Institute of Technology · 2026

Abstract

SenseHuman is a proposed decision support system that combines patient journal entries with wearable sensor data to make everyday health experiences more visible in clinical care. This report presents a user study designed to examine how such a system could be adopted and used in practice. We define the primary user group as patients with ongoing or recurring symptoms, together with healthcare professionals as a secondary user group. The study draws on two theoretical perspectives offered in the course: domestication theory, which explains how a technology becomes part of everyday life, and participatory design, which argues that users should help shape the system rather than only being studied. We selected a mixed-methods approach, with a particular focus on a semi-structured survey, and present a full data collection plan. Preliminary results from eleven respondents (nine patients and two healthcare professionals) indicate strong perceived usefulness (patient mean 4.2 of 5), moderate comfort with data sharing, and clear concerns around effort and privacy. Difficulty in describing symptoms to a doctor emerged as a recurring problem that SenseHuman directly addresses, while healthcare professionals emphasised the need for clear, structured summaries that fit into a time-pressured workflow. Based on these findings, we discuss concrete design implications, the study's limitations, and directions for future work, including expanding the sample of healthcare professionals.

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1 Introduction

For many patients living with ongoing health conditions, the time spent with a doctor represents only a small part of their experience with illness. Symptoms such as fatigue, pain, or breathlessness often occur at home or during daily activities, outside of any clinical setting. These experiences are rarely documented in a structured way. When patients try to describe them during a short visit, important details may be lost or difficult to explain clearly. At the same time, doctors typically rely on the information available during these appointments, which may not fully reflect how a patient has been feeling over days or weeks. This disconnect between everyday health experiences and medical knowledge is a recognized challenge in healthcare, especially for patients whose conditions fluctuate over time (Dinh-Le et al., 2019).

Digital technologies such as wearable devices and mobile health applications have created new possibilities for capturing health information outside formal care settings. There are now hundreds of wearable devices on the market that can connect to health record systems, and this number continues to grow (Dinh-Le et al., 2019). However, having access to these tools does not automatically mean that they become useful in practice. Information collected through wearables or apps often remains separate from medical records, and many patients find it difficult to know how to use the information they collect or how to make it relevant to their care (Fernandes Prabhu et al., 2025). The challenge is therefore not only technical but also about how health information becomes meaningful and usable for the people involved.

Healthcare is also gradually shifting toward models that place greater responsibility on patients to monitor and manage their own health between visits. Patients are increasingly expected to track symptoms, follow treatment plans, and communicate changes in their condition to healthcare professionals. This shift creates a need for tools that support patients in this role without adding to their existing responsibilities. For this to work in practice, technology needs to fit into everyday routines and produce information that is useful in a care context. Simply generating more data that neither patients nor professionals can act on does not solve the problem.

SenseHuman is a decision support system that responds to this problem by combining patient journal entries with data from wearable sensors. What makes the system innovative is that it treats both self-reported experience and physiological measurements as equally important sources of information, rather than relying on one alone. Through a mobile application, patients can record how they feel using text or voice, while a wearable device such as a smartwatch collects heart rate and physical activity in the background. These two sources are then considered together, giving a more complete picture of the patient's situation than either would provide on its own. Research suggests that decision support systems drawing on multiple sources can help healthcare professionals make more informed decisions (Elhaddad & Hamam, 2024). In this way, the system aims to make everyday health experiences more visible in patient care, benefiting both patients and the professionals supporting them.

This report presents a user study for SenseHuman. It defines the intended user group, discusses two relevant theoretical perspectives from the course, selects and justifies a methodology, presents a data collection plan, and reports preliminary results from an initial round of survey data. It concludes by discussing what these findings mean for the design of the system.

2 Background

2.1 User Group

The primary users of SenseHuman are patients with ongoing or recurring symptoms that vary over time. This includes individuals with chronic conditions, rehabilitation needs, or other health-related illnesses in which both subjective experiences (i.e., how a patient feels day to day) and physiological measurements are relevant for understanding changes in health status. Health conditions such as cardiovascular disease, diabetes, chronic pain, and post-operative recovery are examples where this combination may be valuable, since symptoms often shift and are not always captured during a short clinical visit. The system is therefore not aimed at one specific diagnosis, but at situations where continuous monitoring and self-reporting together can provide a more complete picture of a patient's condition over time.

Since the system requires a smartphone and a wearable device, users need access to these tools as well as a basic ability to use them. This creates a practical boundary for the user group, as patients need to be able to record journal entries through text or voice and wear the sensor regularly. Access to these tools is not equal across all groups, and socioeconomic background may therefore influence who is realistically able to use the system. No specific upper age limit applies, but the system may be especially relevant for middle-aged and older adults, since chronic conditions are more common in these groups, though this population should not be treated as one uniform category. Some may already be comfortable with digital tools, while others may need more support with setup or understanding how the system works. The system also depends on continued use over time, since the value of combining journal entries with sensor data comes from repeated engagement rather than occasional interaction.

Users also differ in ways that go beyond age or diagnosis. Previous experience with digital tools, willingness to share personal health data, trust in AI-based systems, and confidence in describing one's own symptoms can all affect how a person interacts with the system. For SenseHuman specifically, these differences are particularly relevant because the system relies on users actively contributing personal information over time. A user who does not trust that their data is handled appropriately, or who feels uncertain about how AI interprets their input, may disengage from the system even if the technology itself functions well. These differences therefore need to be considered not only when defining the user group, but also when making decisions about how the system is designed and communicated to users.

It is also important to recognize that SenseHuman involves more than one type of user. Healthcare professionals, including doctors and nurses, are expected to read and act on the outputs the system produces. Their needs differ from those of patients in important ways. While patients need a system that fits into daily life with limited effort, healthcare professionals need outputs that are clear, structured, and reliable enough to inform clinical decisions. This also means that they need to be able to trust the information the system presents, since acting on unclear or poorly structured data in a clinical setting carries real consequences. Understanding both groups is therefore essential, since a system that works well for one but not the other is unlikely to function effectively in practice.

2.2 Theoretical Perspectives

2.2.1 Domestication Theory

A relevant theoretical perspective for this project is domestication theory (Silverstone & Hirsch, 1992), which describes the process by which technology gradually becomes part of life. This theory is not just about starting to use a device; it is about how people give it meaning and adapt it to their daily routines and practices. Since the proposed SenseHuman system is intended to be used regularly, where symptoms are experienced, interpreted, and recorded by the user (not just reviewed at every visit at the doctor's office), domestication theory is highly relevant.

Instead of only asking whether a user accepts or rejects technology, domestication theory examines what happens after a technology is introduced. A central idea is that technologies do not automatically become useful just because they are available or technically functional. They need to be "domesticated", meaning that users must make them part of their ordinary lives. In the case of SenseHuman, this is especially important because the system depends on repeated and meaningful interaction over time. The user needs to wear a sensor regularly, make journal entries when relevant, and gradually come to see the system as something that supports health management rather than as an extra burden. This means that the success of the technology depends not only on the accuracy of the sensor data or the quality of the AI data processing, but also on whether users are able and willing to integrate the system into their everyday practices.

Domestication theory is often described through the processes of appropriation, objectification, incorporation, and conversion (Silverstone, Hirsch, & Morley, 1992; Haddon, 2006). These concepts are helpful when analyzing how SenseHuman may be used. Appropriation concerns how the technology enters the user's life and whether the user sees it as something worth adopting; in this project, that involves whether a patient understands the value of combining self-reported symptoms with wearable sensor data. If the system is seen as too complicated, intrusive, or unnecessary, it may never be meaningfully adopted. Objectification and incorporation concern how technology is placed within daily life, both physically and practically. A wearable device needs to feel comfortable enough to use regularly, and the journaling function needs to be simple enough to fit into daily routines. If symptom reporting takes too much time, feels emotionally demanding, or interrupts other activities, users may stop engaging with it. In contrast, if the system becomes part of a familiar routine (for example after exercise, during moments of discomfort, or at the end of the day), it is more likely to become a stable part of self-care.

The final concept, conversion, is especially interesting for this project because it concerns how private use becomes connected to the outside world. In SenseHuman, experiences that begin in the personal sphere, such as pain, fatigue, stress, or discomfort, are translated into shareable forms through journal entries, sensor data, and AI-generated summaries. This creates a bridge between the patient's private everyday life and the clinical setting. From a domestication perspective, this is important because the technology does not simply collect information; it changes how lived experience is represented and communicated. Overall, domestication theory shifts the focus from technical possibility to everyday practice (Haddon, 2006), which matters here because the main challenge is not only to build a system that combines journal entries and sensor data, but also to understand how such a system can become meaningful, sustainable, and trusted in the daily lives of its users.

2.2.2 Participatory Design and User Involvement

Another highly relevant perspective for this project is participatory design, together with the broader concept of user involvement (Schuler & Namioka, 1993). This perspective is important because SenseHuman is not a technology that can be designed only from a technical point of view. Even if the system is able to collect sensor data, process journal entries, and generate AI-based summaries, this does not automatically mean that it will be meaningful or useful in practice. Since the system is meant to support both patients and healthcare professionals, its value depends on how well it reflects their real needs, routines, concerns, and expectations. Participatory design emphasizes that users should not only be studied from a distance but should also be involved in shaping the design itself.

User involvement can be understood both as a concept and as a method (Spinuzzi, 2005). As a concept, it refers to the idea that people affected by a technology should have some influence over how it is developed. As a method, it includes concrete ways of involving users, such as interviews, workshops, co-design sessions, testing, and feedback processes. In the case of SenseHuman, this perspective is especially useful because the system brings together different forms of knowledge. Patients have firsthand knowledge of their symptoms, routines, and everyday challenges, while healthcare professionals understand what kinds of information are meaningful in clinical decision-making. A participatory design perspective helps recognize that both forms of knowledge are important and that neither should be treated as secondary.

This is particularly relevant because the intended user group is broad. Users may differ greatly in age, diagnosis, digital habits, daily routines, and levels of confidence in using health technologies. What feels simple and motivating for one user may feel stressful, confusing, or intrusive for another. For that reason, it would be risky to design the system based only on assumptions about what users need. Participatory design offers a way to deal with this diversity by involving users directly in discussions about how the system should work, what kind of interaction feels natural, and what kind of outputs are useful. It also helps balance the needs of the two user groups: patients may prioritize simplicity, emotional reassurance, privacy, and low effort, while healthcare professionals may prioritize clear structure, reliability, and clinical relevance. A participatory approach makes it possible to identify these differences early, instead of discovering conflicts only after the system has been developed.

The perspective also shifts the understanding of design from “designing for users” to “designing with users” (Sanders & Stappers, 2008). In healthcare technologies, there is often a tendency to assume that experts can define the problem and then create a solution for others to use. Participatory design challenges this idea by arguing that users themselves often understand aspects of the situation that experts do not fully see. A patient may know when symptom logging feels burdensome, or what kind of reminders feel supportive rather than stressful; a healthcare professional may know what kind of summary fits into a consultation and what kind of AI-generated output would be ignored in practice. These insights are difficult to identify without direct involvement. For SenseHuman, which deals with intimate health experiences, this makes it necessary to treat user perspectives as a central source of knowledge rather than as a final checkpoint. In this sense, participatory design is not only a method for gathering feedback, but a broader way of thinking about how innovation should take place in health technology.

3 Method

3.1 Study Design

The study uses a mixed-methods design that combines qualitative and quantitative approaches (Creswell & Plano Clark, 2018). The full study is planned in four stages: semi-structured interviews, participatory design workshops, a three-week home study, and a post-use survey. This approach is chosen to understand not only how the SenseHuman system functions, but also how it is experienced and used over time. Since the system depends on regular engagement and combines both self-reported experiences and sensor-based data, it is important to study how users interact with it in different contexts. Given the scope of this course project, our data collection so far has focused on a semi-structured survey, while the remaining stages are described as part of the planned methodology.

The first stage consists of semi-structured interviews with patients (as defined in the user group) and healthcare professionals. These interviews are conducted at the beginning of the design process to explore current practices, expectations, and concerns related to health monitoring and communication. Patients are asked how they currently track symptoms, how they understand changes in their condition, and what challenges they face when describing their experiences to healthcare professionals. They are also asked about their attitudes toward using digital tools and sharing personal health data. Healthcare professionals are asked what types of patient-generated data they consider useful, how such data fits into their workflow, and what concerns they may have regarding reliability, time constraints, or interpretation. The semi-structured format allows participants to speak freely while ensuring that important topics are covered. This stage provides a clear overview of user needs, identifies possible barriers, and helps guide the design process from the start.

The second stage involves participatory design workshops where patients and healthcare professionals work together to shape the system. These workshops are conducted in small groups to support discussion, collaboration, and active participation. Participants are given design scenarios to help them reflect on how the system could work in practice, exploring how subjective experiences such as pain, fatigue, or stress can be combined with sensor data in a way that is meaningful and easy to understand. Participants also contribute to decisions about how they would prefer to interact with the system, for example whether journal entries should be written, spoken, or guided by prompts, and how feedback should be presented. By involving both groups, the study aims to balance their different needs and identify practical solutions that work in real settings.

The third stage is a three-week home study where participants use a functional prototype of SenseHuman in their everyday environments. During this period, participants wear a sensor that collects physiological data such as heart rate and activity, while also recording journal entries through a mobile application. The purpose of this stage is to understand how the system fits into daily routines over a longer period of time, rather than only in short or controlled settings. The final stage is a quantitative survey conducted after the home study, evaluating perceived usefulness, ease of use, trust, satisfaction, and emotional responses such as whether the system feels supportive, reassuring, or stressful. These aspects matter because the success of a health technology depends not only on its technical performance, but also on how it is experienced in everyday use.

Ethical considerations are an important part of the study due to the sensitive nature of health data. All participants are informed about how their data will be collected, used, and stored, and informed consent is obtained before participation begins. Participants are free to withdraw at any time. Data is

handled in a way that protects privacy and confidentiality, and special care is taken with personal journal entries. Since the system depends on user trust, maintaining transparency and data protection is essential throughout the study.

3.2 Data Collection Plan

Data will be collected through four main sources: interviews, participatory design workshops, prototype use during the home study, and a post-use survey. Interviews will be audio-recorded and transcribed, provided that participants give consent, and notes will be taken to capture important observations, recurring concerns, and contextual details. Patient interviews focus on symptom tracking, everyday routines, digital health practices, and attitudes toward sharing personal health data, while interviews with healthcare professionals focus on clinical relevance, workflow, trust, and how patient-generated data could support decision-making.

During the participatory design workshops, data will be collected through group discussions, researcher notes, sketches, design exercises, and feedback on possible interface solutions. These materials help identify what participants consider important, confusing, useful, or problematic, and make it possible to compare the needs of patients and healthcare professionals where their expectations differ. During the three-week home study, the prototype collects usage data such as the frequency of journal entries, the type of input used, and patterns of interaction over time, alongside sensor data such as heart rate and activity levels. After the home study, the survey collects structured responses on perceived usefulness, ease of use, trust, satisfaction, emotional response, and willingness to continue using the system.

For the present report, the first round of data was collected through an online semi-structured survey distributed to potential users. The survey is organized into two tracks, one for patients (the general public) and one for healthcare professionals, so that the same instrument can capture both perspectives. The patient track begins with consent questions, then collects information on current health-tracking habits, the difficulty of describing symptoms to a doctor, preferred ways of recording symptoms, comfort with sharing wearable data and with AI-generated summaries, factors that would build or break trust, and an overall rating of the system's perceived usefulness. The healthcare professional track focuses instead on what patient-generated data they currently use, how useful out-of-visit data is for clinical decisions, the barriers to using it, their preferred data formats, the usefulness and trustworthiness of AI summaries, their main concerns, and whether the system could fit into the clinical workflow. The qualitative data is analyzed through thematic analysis to identify recurring patterns in user needs, concerns, and experiences (Braun & Clarke, 2006), while the quantitative survey data is analyzed descriptively, using frequencies, averages, and comparisons between participant groups.

4 Results

This section presents preliminary results from the first round of survey data collection. At the time of writing, the survey had received eleven responses: nine from the patient track and two from healthcare professionals. Because the sample is small (especially small for healthcare professionals), these results should be read as early, indicative findings rather than as statistically representative conclusions. All respondents gave informed consent and confirmed that they understood they could stop participating

at any time. The two user groups answered different question tracks, so their results are reported separately in Sections 4.2 and 4.3 before being drawn together in Section 4.4.

4.1 Participant Characteristics

Of the eleven respondents, nine identified as patients or members of the general public (one of whom described themselves as a person who regularly monitors health data), and two identified as healthcare professionals. Among the patient group, current tracking habits varied: the most commonly tracked data types were glucose, activity or steps, and sleep (each tracked by four respondents), followed by heart rate (three), and pain, fatigue, and blood pressure (one each). One respondent reported not currently monitoring any health data. Tracking was done mainly through wearable devices (five respondents) and mobile apps (five respondents), with one respondent also using a dedicated medical device. Tracking frequency ranged widely, from several times per day to only when feeling unwell, illustrating the diversity of digital health habits described in Section 2.1.

4.2 Patient Perspectives

Four survey items completed by patients used a five-point rating scale. Figure 1 summarizes the mean scores. The clearest result is that patients rated the overall perceived usefulness of SenseHuman highly, with a mean of 4.2 of 5; six of the nine patients rated it 4 or 5. Comfort with sharing wearable data with healthcare professionals was moderate (mean 3.33), as was comfort with an AI system summarizing their health data (mean 3.11). Both of these items showed a wide spread, with individual answers ranging from 1 (not comfortable) to 5 (very comfortable), suggesting that attitudes toward data sharing and AI vary considerably between individuals.

Figure 1. Mean patient ratings across key survey items (n = 9)

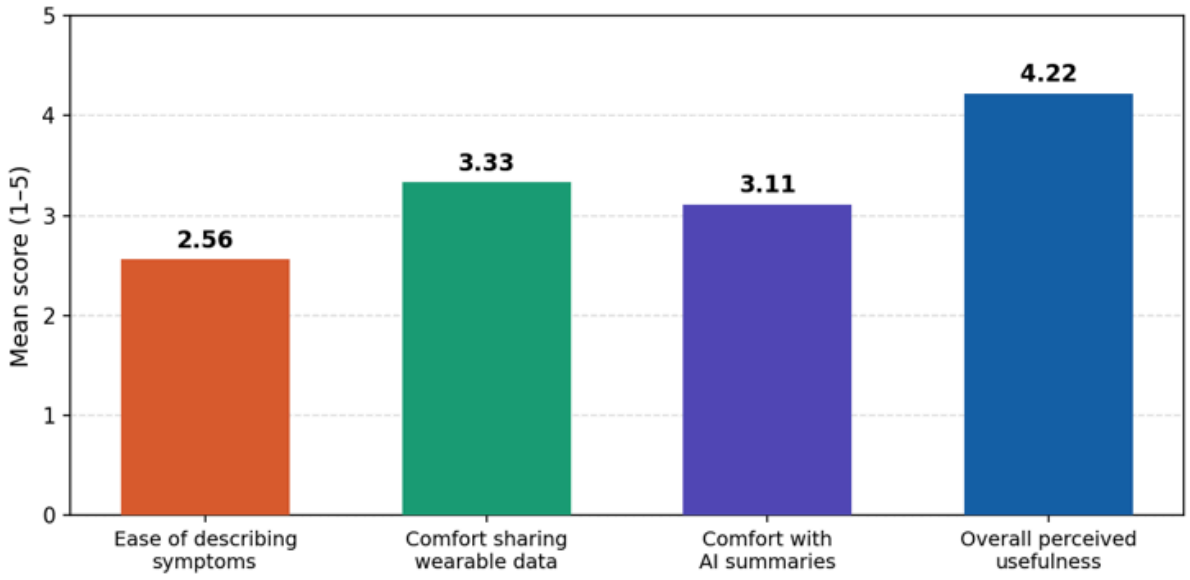


Figure 1. Mean patient ratings across four key survey items (n = 9).

The item on how easy it is to describe symptoms or health changes to a doctor produced the lowest mean score (2.56 of 5), with three of the nine patients selecting the lowest value, indicating that they find this very difficult. This is an important result for SenseHuman, because the difficulty of communicating symptoms is precisely the problem the system is designed to address. When asked

what is most difficult to explain, patients most often selected “what caused symptoms” and “how strong symptoms were” (five respondents each), followed by “when symptoms happened”, “how symptoms changed over time”, and “how symptoms affected daily life” (three each); only one respondent felt that nothing was difficult to explain (Figure 2).

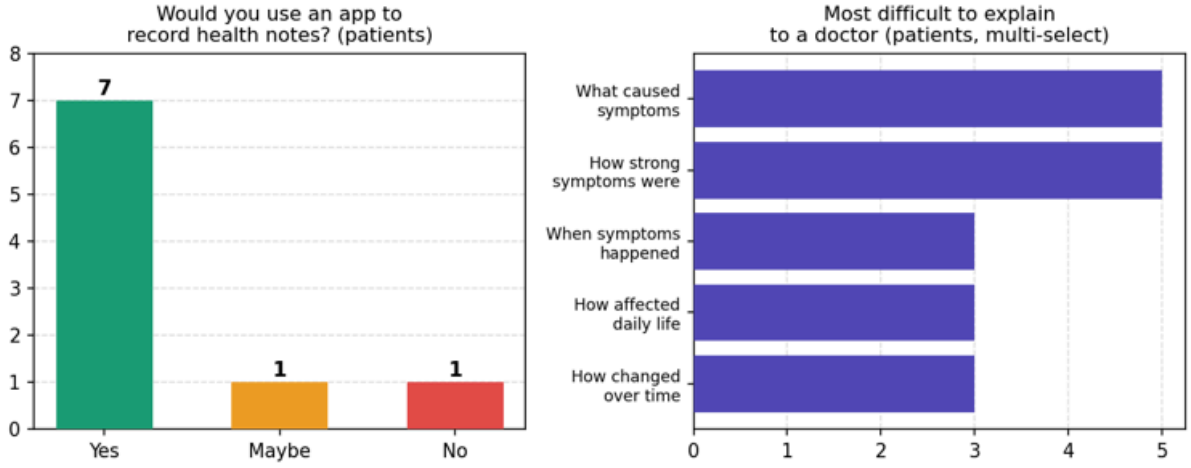


Figure 2. Patient willingness to use the app (left) and aspects most difficult to explain to a doctor (right).

Willingness to adopt the system was generally positive. Seven of the nine patients said they would use an app to record short health notes, one answered “maybe”, and one answered “no”. Eight of the nine were also willing to take part in a follow-up interview, which is encouraging for the planned interview and workshop stages. On qualitative items, trust was clearly conditional: when asked what would make them trust the system, patients most often selected a clear explanation of how data is used and easy-to-understand summaries (three respondents each), followed by the ability for healthcare professionals to review the data (two) and data privacy and security (one). When asked what could make them stop using the system, the most common answer was “too much effort” (three respondents), followed by “difficult to understand” and “technical problems” (two each). Preferences for input format were mixed: voice recording and multiple-choice prompts were each preferred by two respondents, text and rating scales by one each, and three were unsure, which suggests that the system should offer several low-effort input options rather than a single method.

4.3 Healthcare Professional Perspectives

Two healthcare professionals completed the dedicated clinical track of the survey. While this is too small a number to generalise from, their answers give an early indication of clinical priorities and concerns that complement the patient data. Both currently work with some form of patient-generated data (one reported using blood pressure readings and the other a symptom diary) and both rated AI-generated summaries as potentially useful (scores of 5 and 4 of 5). Their views on the overall usefulness of SenseHuman differed, however, with ratings of 3 and 5, and they were split on whether the system could fit into the clinical workflow: one answered “yes” and the other “no”. This split is itself informative, as it suggests that workflow integration, rather than the underlying idea, is the decisive factor for clinical adoption. Table 1 summarizes the healthcare professional responses.

Question	Respondent 1	Respondent 2
Patient data currently used	Blood pressure	Symptom diary
Usefulness of out-of-visit data (1–5)	3	5
Main difficulty using patient data	Lack of time	Unclear structure
Preferred data format	Short summary	Timeline
Usefulness of AI summaries (1–5)	5	4
What makes AI summaries trustworthy	Check original data	Clinician stays responsible
Main concern	Extra workload	Misinterpretation
Could fit into clinical workflow?	No	Yes
Most important for clinical use	Easy access	Clear summaries
Overall usefulness of SenseHuman (1–5)	3	5

Table 1. Responses from the two healthcare professionals (n = 2).

Despite the small number, the two professionals were consistent on several points that matter for design. Both wanted patient data presented in a condensed, structured form rather than as raw data, and both named clarity and easy access as important for clinical use. Their concerns centred on practical barriers rather than the concept itself: extra workload and the risk of misinterpretation. On trust, one emphasised being able to check the original data behind a summary, and the other emphasised that the clinician must remain responsible for decisions. These points align closely with the design priorities identified from the patient data and are reflected in Section 5.

4.4 Cross-cutting Themes

Bringing the two groups together, several recurring themes emerge, and these connect directly to the theoretical perspectives in Section 2.2. First, tracking habits are varied but disconnected from care. Patients track a range of data (steps, glucose, sleep), but this information is rarely brought into conversations with a doctor, and healthcare professionals confirm that out-of-visit data is often hard to use because of limited time and unclear structure. In terms of domestication theory, the data is objectified through devices and apps but not yet converted into a form that is useful in the clinical setting. SenseHuman's core idea, linking self-reported notes to sensor data and summarizing them for a doctor, addresses exactly this gap.

Second, describing symptoms is genuinely difficult for patients, and the specific difficulties identified (what caused symptoms, how strong they were, and when they happened) support the central problem

motivating the system. A structured, time-stamped journal, rather than free recall during a short appointment, could therefore provide real value on both sides of the consultation.

Third, trust and low effort are central, not secondary. Patients named easy-to-understand summaries, clear explanations of data use, and the ability for professionals to review data as the main trust factors, while “too much effort” was the most common reason they would stop using the system. Healthcare professionals echoed this from their side, prioritising clear summaries and easy access while worrying about added workload. From a participatory design perspective, this convergence across both groups is a strong signal that clarity, transparency, and low effort should be treated as core design requirements shaped together with users, not as features added at the end.

5 Discussion

5.1 Design Implications

The design of SenseHuman should prioritize simplicity and low effort for patients. Since the system depends on repeated use over time, journal entries should be quick and flexible. The finding that “too much effort” was the most common reason respondents would stop using the system directly supports this. Users should be able to choose between text, voice, and guided prompts depending on their situation, reflecting the mixed input preferences observed in the results, and the system should avoid making users feel pressured to report constantly, as this may create stress or reduce long-term engagement.

The system should also provide clear explanations of how journal entries and sensor data are used. Since the technology involves sensitive health information and AI-based interpretation, transparency is essential for trust, a point reinforced by respondents who named clear data explanations, professional review and privacy as key trust factors. Patients should be able to see what information has been recorded, understand how it may be shared with healthcare professionals, and have control over their data. Because comfort with data sharing and with AI summaries varied widely in our sample, giving users visible control over what is shared is likely to matter more than assuming a single comfort level for all users.

For healthcare professionals, the design should focus on clarity and clinical usefulness. This was directly reflected in the survey: both professionals wanted patient data in a condensed, structured form (a short summary and a timeline) rather than as raw data, and both named clarity and easy access as important for clinical use. The system should therefore present summaries rather than large amounts of unfiltered data, and important patterns, changes, or patient-reported concerns should be highlighted in a structured way. Their concerns which are extra workload and the risk of misinterpretation, also point to clear requirements: the interface should make it obvious which information comes from the patient, which comes from the sensor, and which has been generated or interpreted by the AI, and it should let a clinician check the original data behind any summary while remaining responsible for decisions. The fact that the two professionals disagreed on whether the system could fit into the clinical workflow reinforces that workflow integration is the decisive factor for clinical adoption and must be designed together with clinicians. A key design implication is therefore that SenseHuman must support both everyday use and clinical interpretation: the patient-facing side should be personal, simple, and reassuring, while the professional-facing side should be structured, reliable, and efficient. Balancing these needs is central to making the system meaningful for both user groups.

5.2 Limitations and Future Work

This study has several limitations. The most important is the small sample: the preliminary results are based on eleven responses, and only two of these came from healthcare professionals which are a key user group whose needs differ substantially from those of patients. While these two responses give a useful early indication of clinical priorities, they are far too few to generalise from, and the patient sample is also modest. The findings should therefore be treated as early indications rather than firm conclusions, and any comparison between the two groups remains tentative. In addition, the survey captures stated attitudes rather than actual behavior; the planned home study with a working prototype is needed to see how the system is really used over time.

The most immediate next step is therefore to substantially expand the sample of healthcare professionals, using the clinical track of the survey that is already in place, and to grow the patient sample so that patterns can be examined more reliably. Beyond this, the planned co-design workshops and three-week home study would allow the project to move from stated preferences to observed use, which is central to both the domestication and participatory design perspectives. In the longer term, future work could explore how SenseHuman might be integrated with a digital twin of the patient, in which journal entries and sensor data continuously update a personalized model of the patient's health to simulate disease progression and predict treatment outcomes. This would extend SenseHuman from a monitoring and communication tool into a more predictive decision support system, but it would also introduce new challenges around data quality, validation, and ethical safeguards. Testing the system with larger and more diverse user groups, and over longer periods, would be necessary to understand whether SenseHuman remains useful after the initial period of use and can become a stable part of everyday health management.

6 Conclusion

This report has presented a user study for SenseHuman, a decision support system that combines patient journal entries with wearable sensor data. We defined a broad primary user group of patients with ongoing or recurring symptoms, together with healthcare professionals as a secondary group, and used domestication theory and participatory design to understand how such a system could become a meaningful and trusted part of everyday life and clinical practice. A mixed-methods approach was selected and justified, and a detailed data collection plan was described.

Preliminary results from eleven respondents suggest that the underlying idea is well received. The nine patients rated the system's potential usefulness highly and confirmed that describing symptoms to a doctor is genuinely difficult, the exact problem the system addresses. The two healthcare professionals, while few, pointed clearly toward the same design priorities from the clinical side: condensed and structured summaries, easy access, transparency about how summaries are produced, and integration into an already time-pressured workflow. Across both groups, the results show that adoption will depend on keeping effort low, giving users control over their data, and making both the input and the AI-generated output easy to understand. These insights, gathered directly from potential users, give the project a clearer and more grounded basis for designing SenseHuman with its users rather than only for them. Expanding the sample of healthcare professionals remains an important next step before firmer conclusions can be drawn.

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